

DECAYING WOODY PLANT MATERIAL AS RESERVOIR OF HUMAN PATHOGENIC OPPORTUNISTIC FUNGI AT JABALPUR

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ABSTRACT: Isolation of opportunistic human pathogenic fungi from decaying woody plant material in Jabalpur by mouse passage technique and direct plating method is presented. A total of 229 isolates representing 24 genera and 69 species were isolated. *Aspergillus* sp. was most frequent (23.5%), followed by *Cladosporium* sp. (15.72%), *Curvularia* sp. (10.91%), *Aureobasidium* (5.6%), *Rhizopus* (5.7%) and *Paecilomyces* (5.2%). In mouse passage technique lungs were found to be a best sources for isolation of opportunistic pathogens. *Curvularia pallescens*, *Cladosporium*, *Trichosporon* sp., *Phialophora* and *Mucor* were isolated from brain. Mouse passage technique has been used very effectively in detecting in presence human fungal pathogens in decaying woody plant materials. Decaying woody plant materials can be a good reservoir and sources of infection for human mycotic diseases.

Key words: *Decaying woody materials, ecology, epidemiology, human pathogenic fungi.*

INTRODUCTION

The opportunistic fungi causing the clinical types hyalo and pheohyphomycosis are increasingly recognized as a cause of life threatening infection in immunocompromised patients (Kwon-Chung and Bennett, 1992). Recently, Matsumoto *et al.* (1994) listed 47 species and 23 genera as agent of hyalohyphomycosis and 57 genera and 101 species of the phaeoid pathogens as the cause of pheohyphomycosis. These causal agents are found in the environment. They are mostly soil fungi associated with plant material as saprophytes or plant pathogens. Out of these decaying tree wood has been reported as reservoir of some important agents of these diseases (Dixon *et al.*, 1980, 1977). In the present study, decaying wood found in the environment of Jabalpur

has been searched for the possible ecological niche of these pathogens so as to determine the possible source of infection in man and animal causing hyalophaeohyphomycosis.

MATERIAL AND METHODS

Total of fifteen samples were collected from woody materials comprising of woody piles, tree stumps, tree trunks and stacked plants. These were collected from different timber and wood dumping sites of Jabalpur. Samples from these randomly selected sites were collected in separate sterilized polythene bags between 1993 to 1995.

Preparation of inoculum from woody materials

About 50 g of each sample was scooped into a clean

polythene bag with the aid of small hand shovel. Mouse passage and direct plating methods were used to examine the samples. Twenty gramme of each sample was transferred to a screw capped sterile conical flasks and the volume was made up to 100 ml with sterile distilled water. The flasks were vigorously shaken for about 15 minutes and allowed to stand further for about an hour. Five ml of supernatant was pipetted out and 0.05mgml^{-1} chloramphenicol was added to inhibit bacterial growth. The supernatant was incubated at 37°C for an hour.

For mouse passage technique

Albino rats four-week-old, disease free, weighing 60g at the time of inoculation was used in the present study. Four animals were housed per polypropylene cage and provided with food and water *ad libitum*. They were maintained in accordance with the guidelines for the care and use of animals. Albino rats were divided into two groups: (1) Test group without cortisone treatment (WCT) and (2) Test group with cortisone treatment (CT). For immunosuppression, the rats were injected subcutaneously aqueous suspension of cortisone acetate (Roussel) in a dose of 125 mg kg^{-1} body weight on day 1, 3 and 5 before injecting the animals with the inoculum suspension.

One ml of supernatant was inoculated intraperitoneally (IP) into each group of animals and were kept under observation for 30 days. Animals either dying or surviving 30 days were sacrificed. All visceral organs viz. lungs, liver, kidneys, heart, spleen and brain were removed aseptically and homogenized using tissue homogenizer in 100 ml sterile saline. The homogenate were cultured on Brain Heart Infusion Agar (BHIA) and Sabourouds Dextrose Agar (SDA) supplemented with 0.05mgml^{-1} chloramphenicol and incubated upto three weeks at 28°C for the recovery of the pathogen from woody materials.

In the direct plating method, one gramme of aseptically ground woody material was mixed with

sterile water (1:10) by shaking vigorously for 5 minutes in a sterile flask. It was then allowed to sediment for 5 minutes. 5 ml of supernatant was kept at 37°C for 1hr. Serial dilution of 1:10, 1:100 and 1:1000 were prepared and 1 ml of each dilution was inoculated in petriplates with BHIA and SDA medium with chloramphenicol ($0.05\mu\text{gml}^{-1}$). All petriplates were incubated at 28°C for four weeks. All visible colonies of pathogenic fungi appearing in cultures were subcultured on SDA slants so as to obtain pure cultures.

RESULTS

A total of 229 fungi belonging to 23 genera and 69 species was isolated both by direct plating and mouse passage technique from decaying woody materials. Table 1 Shows frequency of genera isolated from woody materials. *Aspergillus* sp. was most frequent (23.5%) followed by *Cladosporium* sp. (15.72%) *Curvularia* sp. (10.91%), *Aureobasidium* sp. (5.6%), *Fusarium* sp. (9.8%), *Candida* sp. (9.24%) *Mucor* sp. (8.0%), *Rhizopus* sp. (5.71%) *Paecilomyces* sp. (5.21%) and *Alternaria*. Table 2 shows that 49 fungi were isolated by direct plating method. These belonged to 10 genera and 21 species. *Mucor* sp. and *Aspergillus fumigatus* were most frequently isolated with a frequency of (10.20%). Table 3 shows the isolation of 102 fungi belonging to 18 genera and 59 species by mouse passage technique from cortisone treated (CT) albino rats. The frequency of *Aspergillus* species was maximum. Lungs were the commonest source of isolation of the pathogens followed by spleen, heart, kidneys and liver. *Trichosporon* sp. *Candida* sp., *Cladosporium* sp., *Curvularia pallescens* and *Mucor* sp. were isolated from brain. Table 4 shows isolation of fungi by mouse passage technique from without cortisone treated (WCT) albino rats. These were distributed amongst 17 genera and 32 species. *Aspergillus fumigatus* showed maximum frequency. Kidneys were the best source for the recovery of the pathogens. *Curvularia pallescens* was isolated from brain of WCT albino rats.

Table 1: frequency of isolated genera from decaying woody material

Genera	No. of Isolates	Frequency (%)
<i>Alternaria</i>	11	4.8
<i>Aspergillus</i>	54	23.5
<i>Aureobasidium</i>	13	5.6
<i>Basidiobolus</i>	2	0.87
<i>Canthidia</i>	9	3.93
<i>Chaetomium</i>	2	0.87
<i>Coniobolus</i>	3	1.31
<i>Curvularia</i>	25	10.91
<i>Cladosporium</i>	36	15.71
<i>Calvatrocorpon</i>	2	0.87
<i>Coccidioides</i>	2	0.87
<i>Exophiala</i>	5	2.18
<i>Exserohilum</i>	2	0.87
<i>Fusarium</i>	9	3.39
<i>Mucor</i>	10	4.39
<i>Phialophora</i>	3	1.31
<i>Paecilomyces</i>	10	4.36
<i>Penicillium</i>	7	3.05
<i>Trichosporon</i>	5	1.18
<i>Trichoderma</i>	5	2.18
<i>Rhizomucor</i>	7	3.05
Unidentified fungus	3	1.31
Unidentified fungus	1	1.74

Table 2: Frequency of isolated species from decaying woody material by direct Plating technique using Bhia and Sda Media

Genera	No. of Isolates	Frequency (%)
<i>Aspergillus fumigatus</i>	5	10.20
<i>Aspergillus nidulans</i>	2	5.55
<i>Aspergillus niger</i>	3	8.33
<i>Aspergillus terreus</i>	2	5.55
<i>Aureobasidium pullulans</i>	1	2.77
<i>Aureobasidium sp.</i>	1	2.77
<i>Curvularia lunata</i>	2	5.55
<i>Curvularia pallescens</i>	1	2.77
<i>Curvularia sp.I</i>	3	8.33
<i>Curvularia sp.II</i>	2	5.55
<i>Cladosporium carrionii</i>	1	2.77
<i>Cladosporium sphaerospermum</i>	3	8.33
<i>Cladosporium oxysporum</i>	4	11.11
<i>Mucor sp.</i>	3	8.33
<i>Exophiala jeanselmei</i>	3	8.33
<i>Paecilomyces varioti</i>	1	2.77
<i>Trichosporon sp.</i>	1	2.77
<i>Trichoderma sp.</i>		
<i>Fusarium sp</i>		
Unidentified fungus		

Table 3 : Frequency of isolated species from decaying woody materials by mouse passage technique in cortisone treated albino rats

Name of species	BR	LV	KD	HE	LU	SP	Total no. of isolates	Frequency of species (%)
<i>Alternaria alternata</i>	-	-	+	+	-	-	1	0.98
<i>Alternaria sp. I</i>	-	-	+	-	-	-	1	0.98
<i>Alternaria sp. II</i>	-	+	-	+	+	-	1	0.98
<i>Alternaria sp. III</i>	-	+	-	-	-	-	1	0.98
<i>Aspergillus fumigatus</i>	-	+	+	+	+	+	5	4.90
<i>A. flavus</i>	-	+	+	-	+	-	4	3.92
<i>A. oryzae</i>	-	+	+	-	-	-	2	1.96
<i>A. niger</i>	-	-	-	-	+	+	2	1.96
<i>A. sudowi</i>	-	-	-	+	+	-	1	0.98
<i>A. versicolor</i>	-	-	-	+	+	+	2	1.96
<i>A. terreus</i>	-	+	-	-	+	+	2	1.96
<i>A. ustus</i>	-	+	-	+	-	-	1	0.98
<i>Aspergillus sp.</i>	-	-	-	-	+	-	1	0.98

<i>Aureobasidium</i> sp. I	-	+	-	-	-	3	1.96
<i>Aureobasidium</i> sp. II	-	-	+	+	-	2	1.96
<i>Basidiobolus</i> sp.	-	+	-	-	+	1	0.98
<i>Candida</i> sp. I	-	-	+	+	-	2	1.96
<i>Candida</i> sp. II	+	+	+	+	+	2	1.96
<i>Candida</i> sp. III	+	+	+	+	+	2	1.96
<i>Candida</i> sp. IV	-	-	-	+	-	2	1.96
<i>Cladosporium oxysporum</i>	-	+	+	-	-	1	0.98
<i>C. cladosporioides</i>	-	-	-	-	-	3	2.94
<i>C. carrionii</i>	-	+	-	-	-	4	3.92
<i>C. sphaerospermum</i>	-	-	+	+	+	2	1.96
<i>C. trichoides</i>	-	-	-	+	+	3	2.94
<i>Cladosporium</i> sp. I	+	+	+	-	-	2	3.70
<i>Cladosporium</i> sp. II	-	+	+	-	+	4	3.92
<i>Cladosporium</i> sp. III	-	+	+	-	+	1	0.98
<i>Curvularia lunata</i>	-	+	+	-	-	2	1.96
<i>Curvularia pallescens</i>	+	+	+	+	+	2	1.96
<i>Curvularia geniculata</i>	-	-	+	+	-	1	0.98
<i>Curvularia senegalensis</i>	-	+	+	-	-	1	0.98
<i>Curvularia</i> sp. I	-	-	-	+	-	1	0.98
<i>Curvularia</i> sp. II	-	-	+	-	+	1	0.98
<i>Curvularia</i> sp. III	-	-	+	-	-	1	0.98
<i>Curvularia</i> sp. IV	-	-	-	+	-	1	0.98
<i>Conidiobolus</i> sp.	-	+	-	-	+	3	2.94
<i>Exophiala</i> sp.	-	+	+	-	-	1	0.98
<i>Fusarium oxysporum</i>	-	-	-	+	+	2	1.96
<i>Fusarium solani</i>	-	+	-	-	-	2	1.96
<i>Fusarium</i> sp. I	-	+	-	+	-	2	1.96
<i>Fusarium</i> sp. II	-	-	+	-	+	1	0.98
<i>Fusarium</i> sp. III	-	-	-	+	-	1	0.98
<i>Mucor</i> sp. I	+	-	+	+	-	2	1.96
<i>Mucor</i> sp. II	-	+	-	-	-	1	0.98
<i>Mucor</i> sp. III	+	-	-	-	-	1	0.98
<i>Mucor</i> sp. IV	-	+	-	+	+	2	1.96
<i>Paecilomyces</i> sp.	-	+	+	-	-	2	1.96
<i>Penicillium</i> sp.	-	+	-	-	-	2	1.96
<i>Phialophora</i> sp.	+	+	+	-	-	1	0.98
<i>Rhizomucor</i> sp. I	-	+	-	-	-	1	0.98
<i>Rhizomucor</i> sp. II	-	+	-	+	-	2	1.96
<i>Rhizomucor</i> sp. III	-	-	+	-	-	1	0.98
<i>Trichosporon</i> sp.	+	+	+	+	+	2	1.96
<i>Trichoderma</i> sp.	-	+	+	-	-	2	1.96
<i>Trichoderma</i> sp. I	-	-	+	+	-	1	0.98

BR. = Brain,
KD = Kidney,
LU = Lung,
+ = Positive,

Liv = Liver,
HE = Heart,
SP = Spleen,
- = Negative.

Table 4 : Frequency of isolated species from decaying woody materials by mouse passage technique without cortisone treated albino rats

Name of species	BR	LV	KD	HE	LU	SP	Total no. of isolates	Frequency of species (%)
<i>Alternaria alternata</i>	-	+	+	-	-	-	1	1.16
<i>Alternaria</i> sp.I	-	+	-	-	-	-	3	3.48
<i>Alternaria</i> sp.II	-	-	+	-	-	+	3	3.48
<i>Aspergillus fumigatus</i>	-	+	+	-	-	+	8	9.30
<i>A. terreus</i>	-	+	+	+	+	-	3	3.48
<i>A. niger</i>	-	-	+	+	+	+	6	6.97
<i>A. flavus</i>	-	-	+	+	-	+	2	2.32
<i>Aspergillus</i> sp.	-	-	-	-	+	-	1	1.16
<i>Aureobasidium pullulans</i>	-	-	+	+	-	-	4	4.65
<i>Basidiobolus</i> sp.	-	-	+	-	-	+	3	3.48
<i>Candida</i> sp.I	-	+	+	+	+	+	1	1.16
<i>Candida</i> sp.II	-	-	+	-	+	+	2	2.32
<i>Cladosporium carrionii</i>	-	-	+	-	+	+	2	2.32
<i>Cladosporium sphaerospermum</i>	-	+	+	-	-	+	3	3.48
<i>Cladosporium</i> sp.I	-	-	-	+	+	-	3	3.48
<i>Cladosporium</i> sp.II	-	-	-	-	+	+	4	4.65
<i>Coccidioides</i> sp.	-	+	+	-	-	+	2	2.32
<i>Chaetomium</i> sp.I	-	+	+	-	-	-	2	2.32
<i>Cylindrocarpum tonkenensis</i>	-	+	-	+	-	-	2	2.32
<i>Curcularia lunata</i>	-	+	+	-	-	+	3	3.48
<i>Curcularia senegalensis</i>	-	-	+	-	+	+	4	4.65
<i>Curcularia pallescens</i>	+	-	+	-	+	+	1	1.16
<i>Exophiala jeanselmei</i>	-	-	+	+	-	-	3	3.48
<i>Exserohilum</i> sp.	-	-	+	+	-	+	2	2.32
<i>Paecilomyces varioti</i>	-	+	+	-	-	-	2	2.32
<i>Paecilomyces</i> sp.I	-	-	-	-	-	+	1	1.16
<i>Paecilomyces</i> sp.II	-	-	-	+	-	+	3	3.48
<i>Penicillium</i> sp.I	-	+	+	-	-	-	2	2.32
<i>Penicillium</i> sp.II	-	-	+	-	-	-	3	3.48
<i>Phialophora</i> sp.	-	-	+	+	-	-	2	2.32
<i>Trichosporon</i> sp.	-	-	+	-	-	+	3	3.48
<i>Trichoderma</i> sp.	-	-	+	-	-	+	2	2.32

BR. = Brain,
LU = Lung,

Liv = Liver, KD = Kidney,
SP = Spleen, + = Positive,

HE = Heart,
- = Negative.

In the present investigation, isolation of several genera of hyalo-pheohyphomycosis, medically significant forms of fungal pathogens, from decaying woody plant materials by direct plating technique and mouse passage technique has been presented.

Fifteen samples of decaying plant woody materials from Jabalpur, yielded 229 fungi belonging to 23 genera and 69 species. Out of these, 36 fungi belonging to 10 genera and 21 species were isolated using direct plating technique and 102 fungi were isolated by mouse passage technique, from cortisone treated (CT) albino rats. It is interesting to find that despite the small woody samples examined, this technique has yielded large number of fungi of medical significance. For first time in India very effectively we have used mouse passage technique in detecting human fungal pathogen. Woody materials were found to be the richest sources of several genera of medically significant fungal human pathogens. Such woody materials frequented by man have always been a good source of infection particularly for aspergillosis caused by *Aspergillus fumigatus*, a fungus of great medical significance (Mishra *et al.*, 1992).

Aspergillus species are the most common cause of hospital acquired fungal infection. About 167 species of *Aspergillus* have been reported in literature (Samson 1979, Raper and Fennell, 1965). Out of which 16 species have been documented as etiologic agent of human disease. *Aspergillus fumigatus* is the most common cause of both invasive and non-invasive aspergillosis world wide, Kwon-Chung and Bennett, 1992). Guignani *et al.* (1978) reported *Aspergillus fumigatus*, *A. niger* and *A. flavus* to be associated with mycotic keratitis in Nigeria.

Out of forty species of genus *Curvularia* described in literature, five have been reported to cause infection in humans. These are *Curvularia lunata* (Barde and Singh, 1983, Harris and Dowham 1978), *Curvularia lunata* var. *aeria* (Rinaldi, 1987) *Curvularia geniculata* (Kaufman 1971), *Curvularia verruculosa* (Kwon-Chung and Bennett, 1992, Pouranik 1990, Naidu 1990), *Curvularia pallescens* (Lampert *et al.*, 1977), Agarwal,

1994) and *Curvularia senegalensis* (Laver, 1991). Recently Vishnoi *et al.* (2005) reported *Curvularia geniculata* from the blood of a cancer patient having respiratory problem. Animal studies revealed that *Curvularia geniculata* could develop severe lung infection.

Kaczmarek *et al.* (1986) first isolated *Aureobasidium pullulans* from the blood of a leukemic patient. Salk *et al.* (1986) for the first time reported the ability of *Aureobasidium pullulans* to invade spleen tissues in the patients with disseminated lymphoma. Later, three cases of peritonitis due to *Aureobasidium pullulans* were reported in continuous ambulatory peritoneal dialysis (CAPD) patients (Caporale *et al.*, 1996, Clark *et al.*, 1995 Kaczmarek *et al.*, 1986 Ibanez *et al.*, 1997). Tan *et al.*, (1997) isolated *Aureobasidium pullulans* from BAL suggesting pneumonia in a liver post transplant patient and Wilson *et al.*, (2000) reported that BAL fluid culture from nine patient grew *Aureobasidium* sp. Similarly, *Mucor*, *Paecilomyces*, *Fusarium*, *Penicillium* and *Rhizomucor* are emerging opportunistic human pathogens. Kwon-Chung and Bennett (1992) also reported their isolation in high frequency from decaying woody plant materials that are of considerable medical significance. Similarly, isolation of several mould from decaying wood like *Phialophora* (Singh *et al.*, 1992), *Cylindrocarpum* (Sharma and Chatterjee, 1996) *Paecilomyces* (Uys *et al.*, 1963, Naidu and Singh 1992) *Trichosporon*, (Apaliski *et al.*, 1984), *Chaetomium* (Naidu *et al.*, 1991), *Candida* (Vishnoi, 1998) *Alternaria* (Singh *et al.*, 1990), *Exserohilum* (Agarwal and Singh 1995), and *Basidiobolus* (Chaturvedi, *et al.*, 1984, Koshi, 1972) have been reported to cause infection in man. In the present study, several of these fungi have been isolated from woody material. Thus, woody material could be an important reservoir serving as source of infection in such patients.

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